Kushal Gowda S

Software engineer – ML research, real-time C++, on-device AI and Linux desktop software

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EDUCATION

RV College of Engineering, BE, Computer Science and Engineering · CGPA: 8.93 — 2025 – 2029

PROJECTS CURRENTLY WORKING ON

EmbedEd – contrastive learning for Java code-clone detection — Sep 2026 – present

Python, PyTorch, Hugging Face, Colab · `github.com/ksu-gitreaper807/EmbedEd` (development branch)

- Fine-tuned GraphCodeBERT with a margin-triplet loss on BigCloneBench to test how negative sampling affects clone detection: random negatives reached F1 **0.75** vs **0.25–0.27** for BM25- and embedding-mined hard negatives (6 seeds each).
- Explained the collapse: 99% of embedding-mined negatives already sat closer to the anchor than its true positive, and a blind 100-pair audit (AI-labelled; 95% CIs 19–56%) judged 30–42% of mined "negatives" to be true clones.
- Wrote hypotheses and decision gates before training; 127 offline tests; config fingerprints verified across five Colab VMs.

Real-time photosensitive-epilepsy flash detector (CPPSEYv1) — Feb 2026 – present

C++17, AVX2, FFmpeg, X11/Wayland, PipeWire · `github.com/ksu-gitreaper807/CPPSEYv1`

- Built a real-time screen monitor applying ITU-R BT.1702-3 flash criteria in ~5,100 lines of C++17 (AVX2 kernels, two threaded detectors); a graded risk score drives a dim or blackout response, with X11 and Wayland capture front-ends.
- Found and fixed a false negative that hid every uniform full-screen flash (1–30 Hz) by counting temporal direction flips; in a 33-case synthetic harness **19/19** flash cases now alarm (1/19 before) and 11/12 non-flash controls stay silent.

PROJECTS

GoalGuard – on-device, goal-aware screen-time limiter for Firefox — Sep 2026

WebExtensions MV3, Transformers.js, ONNX · `github.com/ksu-gitreaper807/Non-compete-agreement` (V2 branch)

- Built a Firefox extension that classifies each tab against a weekly goal fully on-device (bundled 33 MB int8 BGE-small embeddings, ~6 ms per title, no cloud calls) and adds timed friction for off-goal pages.
- Designed a layered pipeline (rules, embeddings, optional web context, optional local judge) with graceful fallbacks; 151 unit tests; **77.8%** three-way accuracy and **95.5%** block precision on 153 labelled titles.

WhatsElectron-Linux – unofficial hardened Electron client for WhatsApp Web — Aug 2026

Electron, GitHub Actions · `github.com/ksu-gitreaper807/WhatsElectron-Linux`

- Built a sandboxed Electron app with native Linux integration: notifications grouped per chat with fallbacks, StatusNotifier tray, XDG-portal Do Not Disturb, autostart and a global shortcut (~11.6k lines).
- Wrote a threat model and enforced it: schema-validated IPC with sender authentication, URL and permission allow-lists, CSP; **189** unit tests, strict type-checking, and a CI headless smoke test under Xvfb.

dimsum – compositor-wide dimmer for KDE Plasma 6 — Sep 2026

C++, Qt 6, KWin, OpenGL/GLSL · `github.com/ksu-gitreaper807/dimsum`

- Wrote a KWin 6.7 effect plugin (C++/Qt 6) that renders the Wayland desktop offscreen and redraws it through a GLSL multiply to dim below the hardware backlight minimum; falls back to pass-through on shader failure; shortcuts via KGlobalAccel.
- Wrote build tooling that checks 31 KWin API symbols against the installed headers before every build, plus CMake fallbacks and a six-stage acceptance test script.

COLLABORATIVE WORKS

perfcmp – benchmarking and regression-detection CLI — `github.com/k-karthik-pai/CLI-Tool-For-Relative-Benchmarking`

- C11 command-line tool that compares a baseline and a candidate in paired, randomised rounds and reports the time ratio with an interval, plus scaling and a CI exit code. **Owner:** k-karthik-pai · **Role:** Ideation, thermal cooldown feature

Internship Job Scam Detector – ongoing project — `github.com/k-karthik-pai/Internship-Job-Scam-Detector`

- Ongoing project to detect internship and job scams. **Owner:** k-karthik-pai

SKILLS

Languages: Python, C++17, Bash / Fish

ML & data: PyTorch, Hugging Face (Transformers, Datasets, Hub), Triplet-loss fine-tuning, Embeddings & retrieval (BM25, cosine), Evaluation: F1, MAP@R, bootstrap CIs, Transformers.js · ONNX Runtime Web

Systems & desktop: AVX2 / FMA SIMD, Multithreading, FFmpeg, X11 · Wayland · PipeWire, KWin · Qt 6 · KDE Frameworks 6, OpenGL, Electron, Firefox WebExtensions (MV3), Node.js

Quality & tooling: pytest, Vitest, Node test runner, ESLint · Prettier, GitHub Actions, CMake, Git, Linux (KDE Plasma, Wayland)